

Sebastian Cabrales

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EDUCATION

University of California Berkeley

August 2023 - Present

Bachelor of Arts in Computer Science and Applied Mathematics

Berkeley, CA

Relevant Coursework: CS 189 Machine Learning, Complex Analysis, Data 140 Probability, EECS 127 Optimization, Data 100 Principles of Data Science, Math 110 Abstract Linear Algebra, CS 161 Computer Security, Real Analysis, Abstract Algebra, CS61B Data Structures and Algorithms, CS61C/ Machine Structures

Projects:

Languages: Python, Java, C,

Construction Cost Forecasting & Optimization Platform | Python, IFCOpenShell, AWS Amplify, pandas, Prophet

- Built a cost estimation platform using Python, IFCOpenShell, and React (Vite frontend on AWS Amplify) to parse IFC/Revit files, extract material quantities, and generate project-level cost ranges.
- Developed an optimization engine with time-series analysis (pandas, statsmodels, Prophet) to compute YoY growth, volatility metrics, and forecasts
- Engineered data pipelines linking building materials to relevant BLS PPI series, leveraging pandas for large-scale data handling (~400k NSA observations).

Medical Chatbot with Retrieval-Augmented Generation | Python, Weaviate, Llama Index, Neosync

- Developed a **medical chatbot** utilizing a medical textbook using retrieval-augmented generation.
- Implemented **dynamic paragraph-level chunking** and integrated **Weaviate** as a vector database for efficient context retrieval.
- Developed the front-end chatbot interface using Llama Index, improving user interaction and accessibility.
- Awarded 1st place at the GitHub HQ HackNight for innovative use of Weaviate, Llama Index, and Neosync.

Beauty Advisor | Python, Pinecone, Llama Index, Python

- Designed and implemented a RAG-powered chatbot using Pinecone, Python, and Neosync
- Built a **web-scraping pipeline** with BeautifulSoup and Selenium to collect Sephora articles and beauty forum content, forming a dataset of verified reviews and user opinions.
- Integrated the dataset w/**Pinecone** to enable fast, accurate retrieval of current beauty insights through LlamaIndex and Neosync.

UC Berkeley Cal Teach

June 2024– Present

Computer Science Tutor | Laney Community College

Oakland, CA

- Tutored community college and Data 8 summer students in **Python, Java, data structures, and data analysis**.
- Created practice-exam exercises and led one-on-one/group sessions to improve confidence and performance.
- Supported English language learners with tailored instruction, helping some raise scores to **95%+**.

Educational Development and College Prep Mentor

August 2023 – Present

Oliverhurst, CA

- Guided students from an underfunded high school through **college and scholarship applications** via Zoom.
- Mentored peers as the first CS admit from Olivehurst to UC Berkeley, inspiring others (4 students admitted since 2024).

Awards

Gates Scholar: 1/700 in nation for 2023. Granted Full Scholarship by Bill & Melinda Gates Foundation,

HSF Scholar